

Akash Kothari

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EDUCATION

University of Waterloo

Bachelor of Applied Science in Computer Engineering (Co-op)

Waterloo, ON

GPA: 4.0/4.0

EXPERIENCE

Software Engineer Intern (Incoming)

Sep. 2026 - Dec. 2026

Shopify

Software Engineer Intern

May. 2026 - Present

QuiikMart

Remote

- Built a recommendation engine with a **nightly sweeper** that rolls purchase totals into a baseline popularity store, layered with per-user signals and **Redis** caching for guest sessions.
- Built a bulk CSV product-upload system processing up to **1,000** products per upload, with server-side validation, flexible column aliasing, category matching, and a two-step validate-then-import flow.
- Wrote **80+** unit and integration tests, validated features in staging environments, and shipped bug fixes across a NestJS/TypeScript and PostgreSQL codebase.

ML Software Engineering Intern

Jan. 2026 - Apr. 2026

RamSoft

Toronto, ON

- Built a real-time clinical transcription pipeline running **MedASR** (medical dictation) and **WhisperFlow** (voice commands) in parallel, fusing both streams via timestamp alignment and fuzzy word matching.
- Engineered chunked streaming, PyTorch word-confidence scoring, and a replay simulator for evaluation, cutting error rate by **80%** across **200** clinical audio samples; deployed to **Kubernetes** via Azure Container Registry.
- Designed a **Cosmos DB** → **SQL Server** schema migration for a scheduling epic, with normalized appointment and reservation tables, wall-clock time storage, and full recurrence support.
- Built an internal **MCP server** for Azure SQL with MFA token auth, enabling natural-language database queries.

Cloud Software Engineering Intern

Jun. 2025 - Sep. 2025

Tripleview Technologies

Mississauga, ON

- Automated Teams call-recording retrieval with Azure, Microsoft Graph, and C#, removing **10+ hours/week** of manual work.
- Built an AI summarization pipeline (webhook triggers, polling) that turned **8+ transcripts/hour** into structured **Zendesk** tickets, shipped to production and adopted across internal teams.

PROJECTS

Mercury Mortgages - Expense Automation | React, FastAPI, Supabase, GPT-4o Vision, Redis

- Built a full-stack expense-matching system that matched **600+** receipts to **450+** transactions across **19** sub-companies, eliminating a manual bookkeeping role.
- Engineered a dual-ingestion pipeline parsing **250+** emails via Microsoft Graph webhooks, using **GPT-4o Vision** to extract receipt data and a **Redis** job queue with threaded workers for async PDF processing.
- Built a rules engine mapping vendor and location data to GL codes, with Microsoft SSO and role-based access control.

FocusOS - Personal Productivity OS | FastAPI, PostgreSQL, React, Azure OpenAI, Docker



- Built a full-stack productivity OS with an AI job-analysis pipeline (**Azure OpenAI**) that scores resume-to-job match, extracts ATS keywords, and generates tailored LaTeX resumes.
- Designed a background task pipeline with non-blocking AI calls, retry logic, and status polling on **Docker** and PostgreSQL.

Self-Driving Car Simulator | Unity, C#, Reinforcement Learning

- Built a self-driving car simulation in Unity with custom vehicle physics from scratch, including raycast suspension, tire grip, and braking.
- Implemented a **custom reinforcement learning** training loop (**no Unity ML-Agents**) with **16** raycast sensors and a hand-tuned reward function, reaching a **92%** lap-completion rate within **500** episodes.

TECHNICAL SKILLS

Languages: Python, C/C++, C#, TypeScript, JavaScript, Java, SQL

Frameworks & Libraries: FastAPI, NestJS, Node.js/Express, Flask, React, Next.js, Three.js, Unity

Cloud & DevOps: Azure, Docker, Kubernetes, PostgreSQL, Redis, Supabase, CI/CD, Git, REST APIs

ML & Data: PyTorch, scikit-learn, NumPy, Pandas, Reinforcement Learning, Feature Engineering